b) what if NFA N has ε-arrows?

we define $E(R)$ for $R \subseteq Q$ as the collection of states that we can reach from R by going along ε-arrows, including members of R as well.

$E(R)$ is called ε-closure.

with this updated notation, δ function should be redefined.

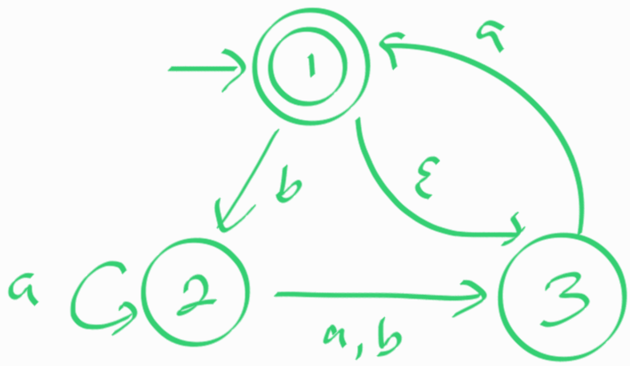
$$\delta'(R, a) = \{ q \in Q \mid q \in E(\delta(r, a)) \text{ for some } r \in R \}$$

Also, we need to change the start state as well.

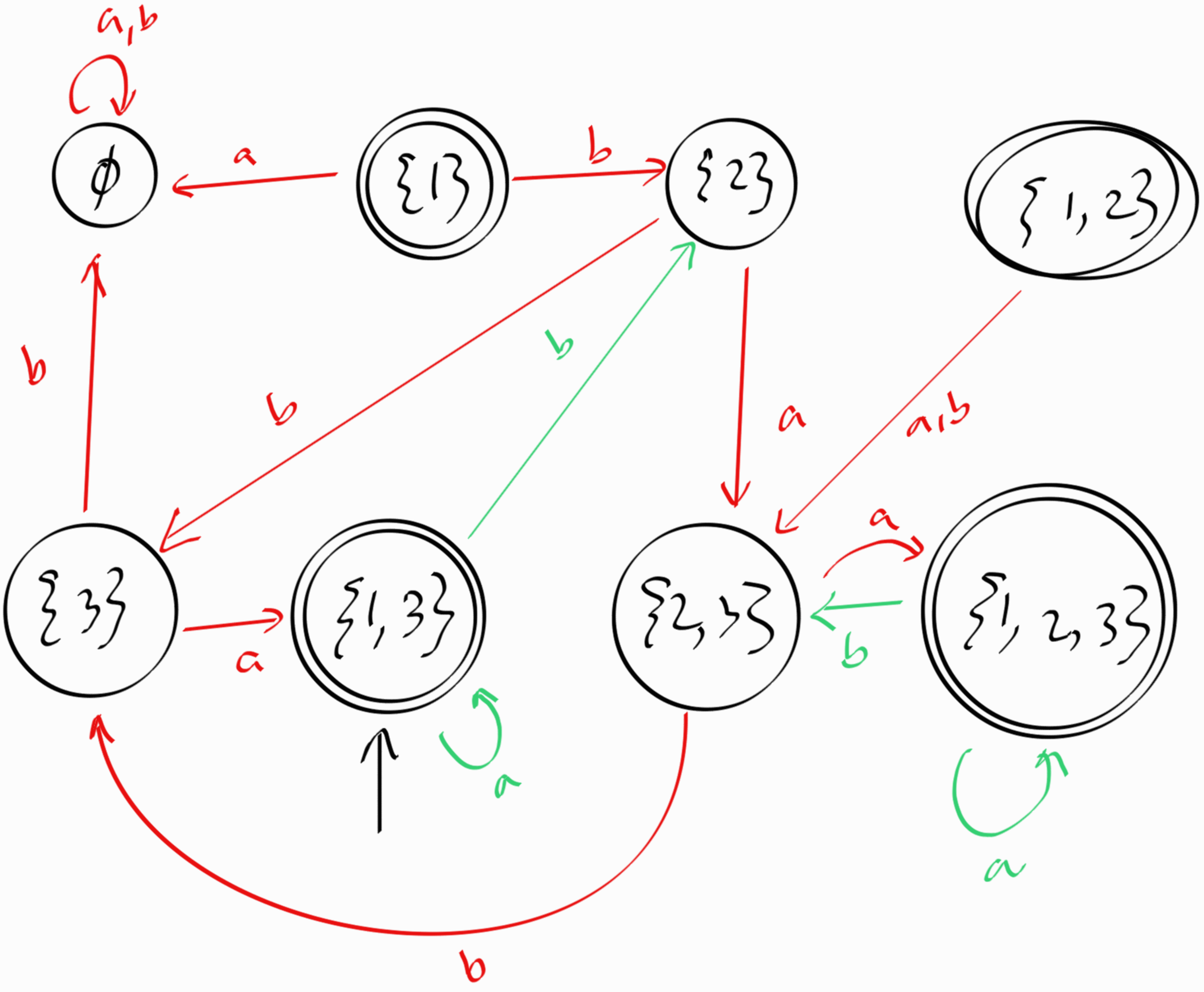
$$q_0' = E(\{q_0\})$$

Let's try to understand $E(R)$

NFA N:



$E(\{1\}) = \{1, 3\}$

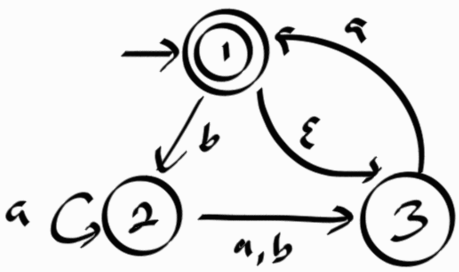


$$\delta'(R,a) = \{ q \in Q \mid E(\delta(r,a)) \text{ for } r \in R \}$$

or

$$\delta'(R,a) = \bigcup_{r \in R} E(\delta(r,a))$$

we will do only some calculations.



$R = \{1,3\}$

$$\delta'(R,a) = \bigcup_{r \in \{1,3\}} E(\delta(r,a)) = E(\delta(1,a)) \cup E(\delta(3,a))$$

$$= E(\emptyset) \cup E(\{1\})$$

$$= \emptyset \cup \{1,3\}$$

$$= \{1,3\}$$

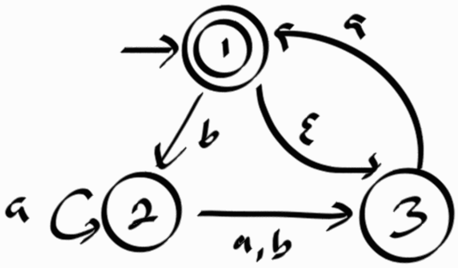
$$\delta'(R,b) = \bigcup_{r \in \{1,3\}} E(\delta(r,b)) = E(\delta(1,b)) \cup E(\delta(3,b))$$

$$= E(\{2\}) \cup E(\emptyset)$$

$$= \{2\} \cup \emptyset$$

$$= \{2\}$$

$R = \{1, 2, 3\}$



$$\delta'(R, a) = \bigcup_{r \in \{1,2,3\}} E(\delta(r,a)) = E(\delta(1,a)) \cup E(\delta(2,a)) \cup E(\delta(3,a))$$

$$= E(\emptyset) \cup E(\{2,3\}) \cup E(\{1\})$$

$$= \emptyset \cup \{2,3\} \cup \{1,3\}$$

$$= \{1,2,3\}$$

$$\delta'(R, b) = \bigcup_{r \in \{1,2,3\}} E(\delta(r,b)) = E(\delta(1,b)) \cup E(\delta(2,b)) \cup E(\delta(3,b))$$

$$= E(\{2\}) \cup E(\{3\}) \cup E(\emptyset)$$

$$= \{2\} \cup \{3\} \cup \emptyset$$

$$= \{2,3\}$$